

Name: _____ Partner: _____ Period: _____

Mission Control — Parametrizing Conic Paths

Mission Control is tracking three objects. Each follows a conic section path described by an implicit equation. Your job: write a parametrization for each object, then use parametric tools to extract key position data.

Object 1 — Space Station (Ellipse)

The station's orbital path satisfies $\frac{(x-1)^2}{64} + \frac{(y+2)^2}{36} = 1$ (units: km).

(a) Identify h , k , a , and b :

$h =$ _____ $k =$ _____ $a =$ _____ $b =$ _____

(b) Write the parametrization ($0 \leq t \leq 2\pi$):

$x(t) =$ _____ $y(t) =$ _____

(c) Find the four extreme positions. Use the parametrization — do not read from a graph.

Rightmost (x max): $t =$ _____ Point: _____

Leftmost (x min): $t =$ _____ Point: _____

Highest (y max): $t =$ _____ Point: _____

Lowest (y min): $t =$ _____ Point: _____

(d) When does the station cross the y -axis? Set $x(t) = 0$ and solve for t , then find the coordinates.

$1 + 8 \cos t = 0 \Rightarrow \cos t =$ _____ $\Rightarrow t =$ _____

y -axis crossing coordinates: _____

Object 2 — Comet (Parabola)

The comet follows the path $y + 3 = 2(x - 1)^2$.

(a) Identify the vertex: _____. Is the parabola vertical or horizontal? _____

(b) Write the parametrization with $x(t) = 1 + t$:

$x(t) = 1 + t$ $y(t) =$ _____ $t \in \mathbb{R}$

(c) When is the comet on the x -axis (i.e., $y = 0$)?

Set $y(t) = 0$: _____

Solve for t : $t =$ _____

Find the x -coordinates: $x =$ _____

Coordinates where comet crosses x -axis: _____

Object 3 — Satellite (Circle)

The satellite follows a circular orbit satisfying $x^2 + y^2 = 100$.

(a) Identify the center and radius:

Center: _____ Radius: _____

(b) Write the standard parametrization:

$x(t) =$ _____ $y(t) =$ _____

Domain for one full counterclockwise orbit: _____

(c) Where is the satellite when $t = \pi/4$?

$x(\pi/4) =$ _____ (use $\cos(\pi/4) = \sqrt{2}/2$)

$y(\pi/4) =$ _____ (use $\sin(\pi/4) = \sqrt{2}/2$)

Coordinates: _____